# LESSON PLAN

## CORRECT AIRCRAFT MAINTENANCE PRACTICES (CAMP) (32)

| | | | |
|---|---|---|---|
| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 04 |
| **Total No. of Lesson Plans** | 10 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

| | |
|---|---|
| **1. Topic** | Construction and navigation of critical aircraft wiring systems |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Construction and navigation of critical aircraft wiring systems |
| **5. Specific Objectives** | 32.1.1.1 Trainees will understand EWIS |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

| | |
|---|---|
| **7. Practical in Lab/Equipment Rooms/Simulators** | Review of wiring schematics and identification of critical system components; Safety: Ensure no live power sources are present; Supervision: Instructor verifies schematic interpretation accuracy |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

| | |
|---|---|
| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) What does the acronym EWIS stand for?
(b) Why is the wiring system considered a critical aircraft system?
(c) What are the primary functions of the aircraft electrical power distribution system?

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • Definition and scope of EWIS<br>• Components of the wiring system<br>• Routing and bundling standards<br>• Identification of critical systems<br>• Navigation of wiring diagrams | L & Ds | PPT, IB | 15 min |
| (b) | • Wiring harness construction methods<br>• Conduit and protective covering types<br>• Termination and connector standards<br>• Grounding and bonding practices<br>• Labeling and identification codes | L & Ds | PPT, PDF | 20 min |
| (c) | • Inspection criteria for critical wiring<br>• Common defects and failure modes<br>• Documentation of wiring issues<br>• Maintenance procedures for EWIS<br>• Safety implications of wiring faults | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain the difference between a standard wiring harness and a critical system wiring bundle.
(b) Describe the steps required to navigate a complex wiring diagram for a critical system.
(c) Identify three common defects that could compromise the integrity of an aircraft wiring system.

| | |
|---|---|
| **12. Home Assignment and Follow Up** | Prepare a brief summary of the key differences between LP 02 and this lesson regarding wiring practices, to be discussed in the next session. |

| | |
|---|---|
| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | Wiring practices and types of inspections | Refer to the master syllabus reference list | LP 05 |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Name, Rank, P No. | Name, Rank, P No. |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min